Breast Cancer Classification using Random Forest & Pipeline

This project demonstrates an **end-to-end machine learning pipeline** for predicting **breast cancer diagnosis** (Benign vs Malignant) using the **Breast Cancer Wisconsin dataset**.

It follows **industrial best practices** by:

- Automating preprocessing with **Pipelines**
- Handling missing values using **`SimpleImputer`**
- Using **Random Forest Classifier** as a robust baseline
- Saving & loading trained models using **Joblib**
- Providing **data visualization** for interpretability

Dependencies

Install the required Python packages before running the project:

```
pip install pandas numpy matplotlib scikit-learn joblib
```

Dataset Information

Source: UCI Machine Learning Repository - Breast Cancer Wisconsin Dataset

Features (10):

1. ClumpThickness
2. UniformityCellSize
3. UniformityCellShape
4. MarginalAdhesion
5. SingleEpithelialCellSize
6. BareNuclei
7. BlandChromatin
8. NormalNucleoli
9. Mitoses

Target:

- **2** = Benign
- **4** = Malignant

Workflow

Data Preparation

- Convert raw `.data` file to `.csv` with **headers**
- Replace **missing values** (`?`) with `NaN`
- Ensure numeric type conversion for all feature columns

Train-Test Split

- Split into **70% train** and **30% test**
- Use **stratified sampling** to preserve class balance

Pipeline Construction

- **Step 1:** `SimpleImputer(strategy="median")` for missing value handling
- **Step 2:** `RandomForestClassifier` with 300 estimators for classification

Model Training & Evaluation

- **Metrics:** Accuracy, Confusion Matrix, Classification Report
- **Feature Importance Plot:** Shows most influential features

Model Saving & Loading

- Save model with **Joblib**
- Load model for future predictions without retraining

Running the Project

Prepare CSV (Only Once)

```
from breast_cancer_pipeline import data_file_to_csv
data_file_to_csv()
```

Train & Evaluate Model

```
python breast_cancer_pipeline.py
```

Expected output:

```
Train Accuracy ::  1.0
Test Accuracy  ::  0.97
Classification Report:
               precision    recall  f1-score   support
...
Confusion Matrix:
[[...]]
Model saved to bc_rf_pipeline.joblib
Loaded model prediction for first test sample: 2
```

Visualizations

- **Feature Importance (Random Forest)**
- **Confusion Matrix** with Matplotlib

Model Storage

- Model is saved as **`bc_rf_pipeline.joblib`**
- Can be loaded anytime for prediction without retraining:

```
from breast_cancer_pipeline import load_model
model = load_model("bc_rf_pipeline.joblib")
```

Sample Prediction

```
sample = test_x.iloc[[0]]
pred = model.predict(sample)
print("Prediction:", pred[0])  # 2 (Benign) or 4 (Malignant)
```

Author

Piyush Manohar Khairnar

Date: 09/08/2025